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Sardar Patel University

[82/A-17] TYBCA -6th Sem (Reg & NC) External Examination-April-2022

US06CBCA23 || Communication and Computer Networks

Date: 05/04/2022, Tuesday

Total Marks:70

Time: 12:00 pm to 2:00 pm

Q. 1 Multiple Choice Questions

[10]

- 1 The data elements sent from one device to another per second is known as _____.
a) Bandwidth b) Modulation Rate c) Data Rate d) None of these
- 2 "Interconnected collection of autonomous computers" is known as _____.
a) Computer network b) Host c) Subnet d) Server
- 3 A _____ cable is made of glass or plastic and transmits signals in the form of light.
a) Twisted-pair b) Coaxial cables c) Fiber Optical d) None of these.
- 4 _____ is an analog multiplexing technique that combines analog signals.
a) FDM b) TDM c) WDM d) None of these.
- 5 In _____ Transmission is use n wires to send n bits at one time.
a) Serial b) Parallel c) Half duplex d) Simplex
- 6 Bit synchronization is responsibility of _____.
a) Physical layer b) Network layer c) Session layer d) Data Link layer
- 7 The signal from a satellite is normally aimed at a specific area called _____.
a) Channel b) Tower c) Orbit d) Footprint
- 8 A Bluetooth Network is called a _____.
a) Piconet b) scatternet c) Wireless d) None of these
- 9 Which of the following topology requires maximum cabling?
a) Bus b) Ring c) Star d) Mesh
- 10 A _____ can be used to solve problem of attenuation.
a) Repeater b) Gateway c) Hub d) Router

Q. 2 Answer the following.

[08]

- 1 _____ network consists of many connection between individual parts of machines.
- 2 In a fiber-optic cable, the signal is propagated along the inner core by _____
- 3 OSI stands for _____.
- 4 A switch performs filtering on the basis of _____ address
- 5 Server consists of two distinct components: transmission lines & switching elements
(True/False)
- 6 Atmosphere is not a guided medium. (True/False)
- 7 A set of layers and protocols is called Protocol stack. (True/False)
- 8 CSMA stand for Carrier Sensitive Multimedia Access. (True/False)

Q. 3 Answer the following. (Attempt any 10)

[20]

- 1 Explain SUBNET in brief.
- 2 List all categories of computer network.
- 3 Define: Data Rate and Modulation Rate.
- 4 Explain serial transmission in short
- 5 Differentiate between TDM and FDM
- 6 List various categories of Coaxial cable and state their use.
- 7 List main responsibilities of Application layer.
- 8 Explain orbit in brief.
- 9 What is satellite? List categories of satellite.
- 10 Explain Gateway in short
- 11 List out advantages of Bus Topology.
- 12 Explain Modem & Repeater in brief.

Q. 4 Answer the following (Attempt any 4)

[32]

- 1 What are the uses/advantages & Disadvantages of computer networks? Explain in detail.
- 2 Explain LAN and WAN in detail.
- 3 Write the difference between circuit switching and packet switching.
- 4 Explain fiber optic in detail
- 5 List all layers of OSI Model and explain any two of them in detail.
- 6 Explain categories of satellite in detail.
- 7 Write a short note on a CSMA.
- 8 List all the topology and explain Any two with their uses, advantages and disadvantages.

X